

Sangram Kumar Sahoo

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EDUCATION

MS in Mechanical engineering - Carnegie Mellon University (CMU)

Expected Graduation: May 2026 | GPA 4.0/4.0

Subjects: MLAI for Engineers, Modern Control Theory, Soft Robots, Intro to Deep Learning, Engineering a Startup

BTech in Mechanical Engineering – SRM Institute of Science and Technology, Chennai

Graduated: May 2023 | GPA: 8.7/10

PROJECTS

Detection of water purity level using an underwater drone system

(Aug - Dec 2024)

- The technical problem this project aims to develop a bio-inspired soft robotic device that monitors water quality and detection of potential of hydrogen in the water bodies, overcoming the limitations of stationary sensors and manual sampling methods.
- The soft robotic structures impact maneuverability and how sensor placement affects data accuracy in dynamic water conditions.

Reinforcement Learning-based neural architecture search for deepfake voice detection

(Jan - May 2025)

- Established a comprehensive data processing pipeline for audio preprocessing and feature extraction
- Implemented and evaluated baseline models achieving an Equal Error Rate (EER) of 2.8% with 50 epochs
- Developed an initial proof-of-concept for the Neural Architecture Search framework (currently under refinement)

PROFESSIONAL EXPERIENCE

Computational Engineering and Robotics Lab (CERLAB)

CMU, Pittsburgh

Graduate Researcher - RoboConstruction Team.

(Aug 2024 -May 2025)

- Developed an Isaac Sim simulation environment for SLAM test bed with the ultimate goal of developing a SLAM algorithm for mining environment.
- Simulated different LiDARs and camera integration for simple hauling and loading operation for the wheel loader in a simulated environment.

Graduate Researcher - Shapeanalysis Team.

(May 2025 -Present)

- Working on a graph neural net on engineering drawing for classification and image marking on different drawings.
- Studied different research papers on the relevant topic for this project.

Kansai Nerolac Paints

Hosur, India

Mechanical Engineering Intern

(Jun - Jun 2022)

- Completed the daily production checklist and generated comprehensive reports on daily activities. Operated industrial machinery and acquired a thorough understanding of various manufacturing processes. Assisted in maintaining the ball mill and the daily routine of the RO water system within the manufacturing plant.

Youth India E-school

Bhubaneswar, India

Deputy General Manager

(Dec - July 2024)

- Oversee all activities pertaining to operational metrics, quality standards, and human resources management. Oversee and train four sales associates, two team leaders, and two support managers. Provide training on customer service and conflict resolution techniques to employees.

RESEARCH EXPERIENCE

Design of pedicle screw fixation for lumbar spine

(Jan – May 2023)

SRM Institute of science and technology

- Developed a FE model to study pedicle screw fixation and the effect of screws by varying the size and material. The distribution of stress on different screws with variation of forces applied on the vertebrae was observed.

SKILLS

Frameworks: Python (PyTorch, TensorFlow, CUDA, PySpark, OpenCV), C++ Docker, ROS, Kotlin.

Database & Cloud: Google Cloud Platform (GCP), AWS, Power BI.

Softwares: Solidworks, Autocad, Matlab

LEADERSHIP

- **Youth India Foundation**, Co-Founder

Feb 2020 – Present